Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Zhong, Fang Ph.D.		
Project Title:	Measurement of the Thermal Conductivity Near the Liquid-VaporCritical Point of He-3 and He-4		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	02/01/2001	**End Date:**	05/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	High Resolution (0.5nK) Thermometry (HRT) will be used to measure the thermal conductivity of fluid He-3 and He-4 near its respective liquid-vapor critical point. Our calculation shows that the use of HRT will reduce density stratification by an order of 1000 times compared to the use of conventional germanium thermometry for temperature measurements. The combination of using HRT and microgravity will enable us to measure thermal conductivity at least 100 times closer to the critical temperature than any ground-based experimental measurements. It is in this extended region where the thermodynamic properties of fluid He-3 and He-4 exhibit asymptotic, nearly pure, power-law divergences. 1) Measure the effective thermal conductivity of He-3 and He-4 over a wide range of reduced temperatures around its respective critical temperature. 2) Compare the experimental results to the recent predictions of both the Mode-Coupling theory and the Renormalization Group theory.		

	3) Determine the feasibility of space-based, high-resolution thermal conductivity measurements, and examine the possibility that such experiments will significantly improve upon earth-based measurements. The thermal conductivity and shear viscosity of a simple fluid diverge near its liquid-vapor critical point as a result of the diverging correlation length. Dynamic scaling theory predicts a scaling relation for the critical exponents of the two dynamic properties. This proposed experiment will provide high quality data for the comparison to the theory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)